

August 6, 2026

Deloitte Canada

Dear Hiring Team,

I am writing to apply for the AI Research Engineer position at Deloitte. I am a recent M.Sc. Computer Science graduate from the University of Calgary with applied research and software experience in Python, PyTorch, model evaluation, large-scale data processing, and translating research ideas into reliable working systems.

My graduate work focused on machine learning and data systems for complex Earth observation data. In one project, I trained and evaluated deep ResNet models for digital elevation model super-resolution, built reproducible PyTorch pipelines, and compared validation strategies to avoid overly optimistic performance conclusions. That experience maps well to Deloitte's emphasis on building evaluation frameworks, measuring model behavior, and making AI systems trustworthy enough for real use.

I also developed scalable C++ and Python software for spatiotemporal satellite-data management using DGGS, wavelets, and tensor-based storage. The work required careful experimental design, performance analysis, documentation, and communication through a first-author Remote Sensing publication and my M.Sc. thesis. Alongside this research, I have built backend services, supported AWS-based data-pipeline coursework, and recently mentored students through Neuromatch Academy's deep learning program, including PyTorch tutorials on attention, transformers, diffusion models, and reinforcement learning.

I am especially interested in Deloitte because the role sits at the intersection of applied AI research, responsible deployment, and product-focused engineering. I can contribute strong foundations in Python, deep learning, reproducible experimentation, and technical communication, while continuing to grow in production GenAI tooling such as advanced LLM fine-tuning, inference optimization, and agent frameworks. I would be excited to help design, evaluate, and document AI systems that are practical, reliable, and aligned with responsible AI principles.

Thank you for considering my application. I would welcome the opportunity to discuss how my research background, engineering experience, and interest in applied GenAI systems could contribute to Deloitte's AI and Data practice.

Sincerely,
Amir Mirzai Golpayegani
amirgolpaa24@gmail.com